Ch. 14 Solutions

Principles of Microeconomics

July 28, 2026

Quick Quiz

1. (c) – His explicit cost is $100 for seeds. His implicit cost is $200 from the income he forewent by not teaching banjo lessons.

2. (a) – His explicit cost is $10 for ingredients. His implicit cost is $40 from the income he forewent by not mowing his neighbor's lawn. That means his accounting profit is $60 - 10 = 50$, and his economic profit is $60 - (10 + 40) = 10$.

3. (a) – She got two additional bushels from her second seed. Since the seeds have diminishing marginal productivity, she will get fewer than two additional bushels from her third seed. Therefore, she must get fewer than 7 bushels from 3 bags.

4. (d) – The production function gets flatter because each additional unit of input increases output by less than the last unit of input. The total cost curve gets steeper because each additional unit of output requires the firm to increase its inputs by more than the last unit of output did.

5. (d) – Increasing output by one costs $8, so marginal cost is $8. $5,008 divided by 1,000 units is roughly $5, so average total cost is $5.

6. (c) – Since marginal cost is less than average total cost, producing one more unit will pull average total cost down.

7. (b) – The license fee is a fixed cost because it doesn't depend on how much output the firm produces. Fixed costs factor into the average total cost and the average fixed cost curves.

8. (a) – Specialization makes workers more productive, so the firm's marginal product of labor increases. If the marginal product of labor increases, that means the firm's marginal cost will fall which will in turn drag down average total cost. If average total cost falls, the firm is exhibiting economies of scale.

9. (d) – As production increases, average total cost increases. That means Boeing exhibits diseconomies of scale.

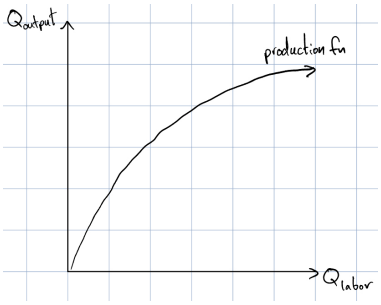
Questions for Review

1. Profit = Total Revenue - Total Cost

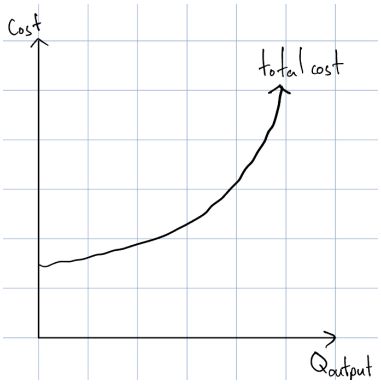
2. The money that firms spend establishing their business could earn interest if the firm saved it instead. We call the interest that they would have earned the cost of capital. Since the cost of capital in this case is an implicit cost – it doesn't require a cash outlay – accountants would ignore it.

3. The marginal product of an input is the change in output from a one unit increase in that input. Diminishing marginal product is the idea that each additional unit of an input increases output by less than the last unit of that input did.

4. The production function is increasing at a decreasing rate because each additional unit of labor increases output by less than the last unit of labor did.



The total cost curve is increasing at an increasing rate because each additional unit of output requires hiring more additional labor than the last unit did. That makes each additional unit of output more expensive to produce than the last.

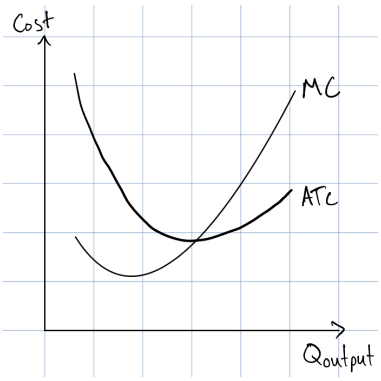


5. Total cost is how much the firm has to spend on inputs to produce all of its output. Mathematically, it's the sum of the firm's explicit and implicit costs. Alternatively, it's the sum of the firm's fixed and variable costs.

   Average total cost is the cost of producing a typical unit of output. Mathematically, it's total cost divided by total output.

   Marginal cost is the cost of producing an additional unit of output. Mathematically, it's the change in total cost divided by the change in total output.

6.



As output increases, marginal cost typically decreases at first before eventually increasing. That's because the marginal product of labor typically increases at first which drives down marginal cost, but then eventually it decreases which drives up marginal cost.

Average total cost has a similar U-shape. It decreases at first because average total cost equals the sum of average fixed cost and average variable cost, and average fixed cost falls rapidly at fist as the fixed costs are spread across the first few units of output. Eventually, average total cost increases because diminishing marginal product causes average variable costs to increase, and the increase in average variable cost overwhelms the decrease in average fixed cost.

The marginal cost curve intersects the average total cost curve at its minimum. If marginal cost is below average total cost, then producing an additional unit of output will pull the average total cost curve down. If marginal cost is above average total cost, then producing an additional unit of output will pull the average total cost curve up. In other words, if marginal cost is below average total cost, average total cost is decreasing, and if marginal cost is above average total cost, then average total cost is increasing. That means the average total cost curve flips from decreasing to increasing when the marginal cost curve moves from below it to above it. Therefore, the marginal cost curve intersects average total cost curve at its minimum.

7. In the long run, firms' ATC curves are a flatter U-shape and lie below its short run ATC curves. That's because in the long run there are no fixed costs – all costs are variable. As a result, firms have more flexibility to choose their inputs and keep their costs low.

8. A company exhibits economies of scale if its average total costs decline as its output increases. Economies of scale often occur when employing more labor allows for greater specialization and therefore greater productivity.

   A company exhibits diseconomies of scale if its average total costs increase as its output increases. Diseconomies of scale often occur when companies get so large that there are coordination problems that make it less productive.

Problems and Applications

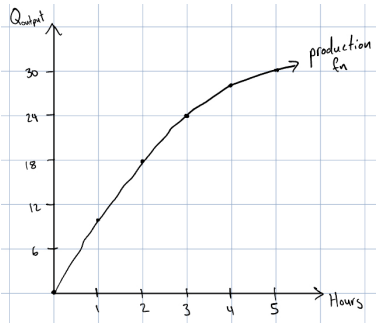
1. (a) opportunity cost
   (b) average total cost
   (c) fixed cost
   (d) variable cost
   (e) total cost
   (f) marginal cost

2. (a) The opportunity cost of a choice is what you give up to make it.
   (b) $\$350,000 + \$80,000 = \$430,000$
   (c) $\$400,000 - \$350,000 = \$50,000$
   (d) She should not. Her economic profit is negative because her opportunity cost is greater than her total revenue. In other words, she would be better off if she remained a vampire hunter.
   (e) It would need to generate more than her opportunity cost, $430,000.

3. (a)

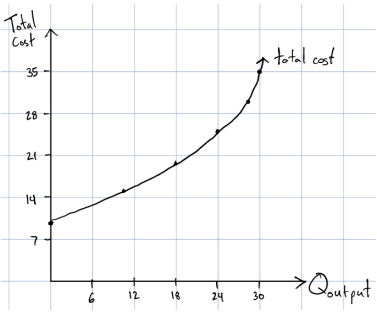
Hours	**Quantity of Fish (in pounds)**	**Marginal Product**
0	0	–
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2

(b) The production function is increasing at a decreasing rate because the fisherman's marginal product of time is diminishing. The more hours he works, the less productive each additional hour is.



(c) The total cost curve is increasing at an increasing rate. Because the fisherman's marginal product of time is diminishing, each additional fish that he catches takes him more time to catch than the last one. That means his opportunity cost accelerates as he catches more fish.

Hours	**Quantity of Fish (in pounds)**	**Fixed Cost**	**Variable Cost**	**Total Cost**
0	0	10	0	10
1	10	10	5	15
2	18	10	10	20
3	24	10	15	25
4	28	10	20	30
5	30	10	25	35



4.

Workers	Output	Marginal Product	Total Cost	Average Total Cost	Marginal Cost
0	0	–	200	–	–
1	20	20	300	15	5
2	50	30	400	8	3.33
3	90	40	500	5.56	2.5
4	120	30	600	5	3.33
5	140	20	700	5	5
6	150	10	800	5.33	10
7	155	5	900	5.81	20

(a) At first, marginal product is increasing. Then, it diminishes. Additional workers might be increasingly productive at first because having multiple workers allows each worker to specialize and become better at their particular task. Eventually, additional workers might become decreasingly

productive because there isn't enough work for each worker to operate at maximum capacity or because it gets harder to coordinate a larger work force.

(b) See above.

(c) Average total cost decreases at first and increases later.

(d) Marginal cost decreases at first and increases later.

(e) Marginal product increases at first and decreases later. Marginal cost decreases at first and increases later. The trend in marginal product explains the trend in marginal cost. Since workers get increasingly productive at first, it requires less additional labor to produce an additional unit of output. Less additional labor means lower marginal costs. When workers get less productive, it requires more additional labor to produce an additional unit of output. Hiring more additional labor means higher marginal costs.

(f) When marginal cost is lower than average total cost, average total cost is decreasing because producing additional units at lower than average cost pulls the average down. When marginal cost is higher than average total cost, average total cost is increasing because producing additional units at higher than average cost pulls the average up. Marginal cost intersects average total cost at average total cost's minimum because that's where average total cost switches from decreasing to increasing.

5. No, I shouldn't. At 600 consoles, total cost is $600 * \$300 = \$180,000$. At 601 consoles, total cost is $601 * \$301 = \$180,901$. That means the marginal cost of producing the 601st console is \$901. Therefore, producing another console would cost me more than the person is offering.

6. (a) Its fixed cost is \$300.

(b) The change in total cost equals the change in variable cost. That's because fixed cost doesn't change when the quantity of output changes, so the change in total cost is entirely the change in variable cost. Mathematically,

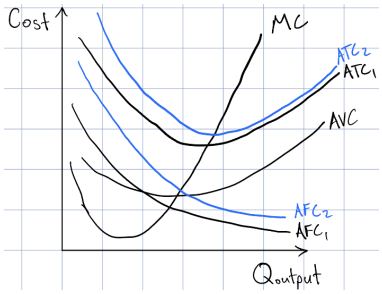
$$\begin{aligned}\Delta TC &= TC_2 - TC_1 \\ &= (FC + VC_2) - (FC + VC_1) \\ &= FC + VC_2 - FC - VC_1 \\ &= VC_2 - VC_1 \\ &= \Delta VC\end{aligned}$$

Quantity	TC	VC	$MC = \Delta TC$	$MC = \Delta VC$
0 dozen pizzas	\$300	\$0	–	–
1	350	50	50	50
2	390	90	40	40
3	420	120	30	30
4	450	150	30	30
5	490	190	40	40
6	540	240	50	50

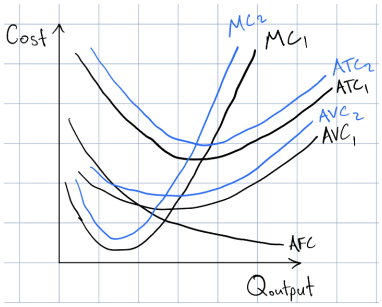
7. The efficient scale of the company is 4 houses painted per month.

Quantity of Houses Painted per Month	1	2	3	4	5	6	7
Variable Costs	\$10	\$20	\$40	\$80	\$160	\$320	\$640
$AFC = \frac{FC}{Q}$	200	100	66.67	50	40	33.33	28.57
$AVC = \frac{VC}{Q}$	10	10	13.33	20	32	53.33	91.43
ATC	210	110	80	70	72	86.67	120

8. (a) A lump-sum tax is a single payment that every person or firm has to pay regardless of their circumstances. Since it doesn't depend on the quantity of output produced, it's a fixed cost. Therefore, the average fixed cost and average total cost curves shift. Marginal cost doesn't shift because marginal cost just depends on variable cost (as we saw in 4).



(b) The per-burger tax is a variable cost, so the average variable cost, average total cost, and marginal cost curves shift.

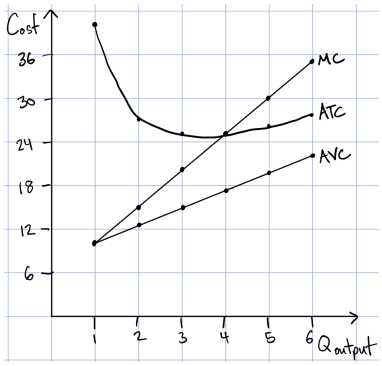


9. (a)

Quantity	VC	TC	AVC	ATC	MC
0	0	30	–	–	–
1	10	40	10	40	10
2	25	55	12.5	27.5	15
3	45	75	15	25	20
4	70	100	17.5	25	25
5	100	130	20	26	30
6	135	165	22.5	27.5	35

(b) When MC is less than ATC, producing one more unit pulls ATC down, so ATC is decreasing. Similarly, when MC is greater than ATC, producing one more unit pulls ATC up, so ATC is increasing. Therefore, when MC intersects ATC, ATC switches from decreasing to increasing, so ATC is at a minimum.

When MC is greater than AVC, producing one more unit pulls AVC up, so AVC is increasing. It intersects with AVC when quantity is one because at that quantity marginal cost equals variable cost, and the denominator of average variable cost is one.



10. Below is a table of each firm's ATC.

Quantity	1	2	3	4	5	6	7
Firm A	\$60	\$35	\$26.67	\$22.5	\$20	\$18.33	\$17.14
Firm B	11	12	13	14	15	16	17
Firm C	21	17	16.33	16.5	17	17.67	18.43

Firm A experiences economies of scale because their ATC decreases as their output increases. Firm B experiences diseconomies of scale because their ATC increases as their output increases. Firm C experiences economies of scale for low levels of output and diseconomies of scale for high levels of output.

Non-Textbook Questions

1. (a)
$$ATC = \frac{\$40}{4} = \$10$$

(b)
$$ATC = \frac{\$40 + \$5}{5} = \$9$$

My average total cost decreased because the marginal cost of producing another widget was less than my initial average total cost. As a result, producing that widget pulled my average total cost down.

(c)
$$ATC = \frac{\$40 + \$20}{5} = \$12$$

My average total cost increased because the marginal cost of producing another widget was greater than my initial average total cost. As a result, producing that widget pulled my average total cost up.

2. (a) Total cost is fixed cost plus variable cost.

	TC		
Quantity of trucks	**20 orders**	**40 orders**	**60 orders**
2	\$8,000	\$11,000	\$18,000
3	8,800	10,800	17,800
4	9,200	11,600	16,400

(b) 20: He should purchase 2 trucks because that will minimize his total cost. His average total cost will be $\$8,000/20 = \$400$.

40: He should purchase 3 trucks because that will minimize his total cost. His average total cost will be $\$10,800/40 = \$270$.

60: He should purchase 4 trucks because that will minimize his total cost. His average total cost will be $\$16,400/60 = \$273.33$

(c) After business declines, his average total cost is $\$8,800/20 = \$440$ in the short run.
If business booms, his average total cost will be $\$17,800/60 = \$296.67$

(d) For 20 orders per week, Don's long-run average total cost is \$400 because in the long run he will minimize costs by selling a truck so that he only has two.
His long-run average total cost is lower than his short-run average total cost because in the long run he has the flexibility to manage the number of trucks he owns. His fixed costs become variable, and that allows him to further minimize his costs.

3. (a) $\$1,000(0.05) = \$50$.

(b) Her marginal cost increases when the opportunity cost of her time increases.

$$MC = \begin{cases} \$6 + \frac{1}{2}(\$15) & Q <= 1,500(2) \\ \$6 + \frac{1}{2}(\$24) & Q > 1,500(2) \end{cases}$$

$$= \begin{cases} \$13.5 & Q <= 3,000 \\ \$18 & Q > 3,000 \end{cases}$$

(c) She should produce 3,000 sets because the marginal cost of producing the first 3,000 sets is less than the \$15 of revenue she'd make by selling them. The marginal cost of producing any sets beyond 3,000 is greater than the revenue she'd make by selling them, so she shouldn't make any more than 3,000.

(d) $EC = \$1,000 + \$6(3,000) = \$19,000$

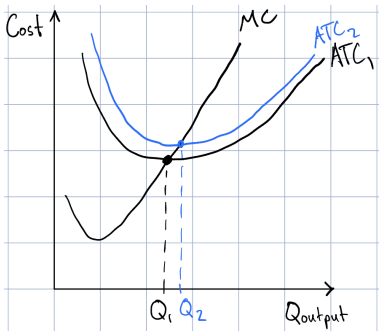
(e)
$$IC = \$1,000(0.05) + \$15\left(\frac{3,000}{2}\right) = \$50 + \$15(1,500) = \$50 + \$22,500 = \$22,550$$

(f)
$$\text{Economic Profit} = \$15(3,000) - (\$19,000 + \$22,550) = \$45,000 - \$41,550 = \$3,450$$

(g)
$$\text{Accounting Profit} = \$45,000 - \$19,000 = \$26,000$$

4. (a) True – If marginal product is decreasing, that means the input is getting less productive as the firm uses more of it. That means if the firm wants to produce an additional unit of output, it will have to use more additional inputs than it did to produce the last unit of output. That's more expensive, so the marginal cost must be rising.

(b) True – If fixed cost increases, total cost increases, and marginal cost remains unaffected. That means the average total cost curve shifts up while the marginal cost curve doesn't move. Since marginal cost is increasing, average total cost must intersect marginal cost at a higher quantity after the shift. That quantity is the efficient scale of the firm, so the efficient scale of the firm increases.



(c) False – An increase in fixed cost *does not* increase marginal cost.

(d) False – When marginal cost is above average total cost, average total cost must be *rising*.